

# Lab-04 Data Science Hands-on Guide

## Overview

This is a hands-on lab that demonstrates the end-to-end ML pipeline for forecasting daily stock inventory levels across multiple SKUs using Azure Databricks. Followed by how to invoke the **inventory-forecast-endpoint** using an AI agent connected via the **Model Context Protocol (MCP)**.

**The goal is to provide a hands-on experience in creating an end-to-end working agentic solution using a custom created ML model in Azure Databricks.**

You will complete the following steps during the session.

| Step | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 1    | <b>Configuration</b> — Set experiment directory and model names                                     |
| 2    | <b>Synthetic Data</b> — Generate daily inventory for 5 SKUs with trend, seasonality & replenishment |
| 3    | <b>EDA</b> — Visualise time series patterns, distribution and seasonal effects                      |
| 4    | <b>Databricks AutoML</b> — Auto-train & tune forecasting models (10-minute budget)                  |
| 5    | <b>Prediction</b> — Load best model, generate 30-day forward forecasts                              |
| 6    | <b>Install packages</b> — databricks-langchain, mcp, fastmcp                                        |
| 7    | <b>Configuration</b> — Endpoint name, workspace URL, LLM endpoint                                   |
| 8    | <b>Define MCP Server</b> — FastMCP server exposing the forecast endpoint as a tool                  |
| 9    | <b>Build Agent</b> — LangGraph ReAct agent connected to the MCP server                              |
| 10   | <b>Test Agent</b> — Invoke forecasts via natural language queries                                   |

## Prerequisite

### Clone Repo

Please clone the following repo: git clone <https://github.com/mufajjul/ai-workshop-infra.git>

## Environment provisioning.

Please run the following script to provision the DB workspace.

```
sh -I infra/Databricks/ARM/cli.sh
```

## Activity 1: Cluster Creation

Please create a cluster per person for this session. Recommendation is to use a **single node cluster** to avoid any sub-level SKU quota issue.

The screenshot shows the 'Create new compute' interface in Databricks. The left sidebar contains navigation options like 'Workspace', 'Recents', 'Catalog', 'Jobs & Pipelines', 'Compute', 'Marketplace', 'SQL', 'SQL Editor', 'Queries', 'Dashboards', 'Genie Spaces', 'Alerts', 'Query History', 'SQL Warehouses', 'Data Engineering', 'Runs', 'Data Ingestion', 'AI/ML', 'Playground', 'Agents', 'AI Gateway', 'Experiments', 'Features', 'Models', and 'Serving'. The main panel is titled 'Create new compute' and has a breadcrumb 'Compute > New compute > Simple form: ON'. The 'General' tab is active, showing a 'Compute name' field with the value 'Mufajjul Ali's Cluster 2026-06-19 12:13:51' and a 'Policy' dropdown set to 'Unrestricted'. The 'Performance' section includes a 'Machine learning' checkbox (unchecked), a 'Databricks runtime' dropdown set to '17.3 LTS' (circled in red), a 'Scala 2.13, Spark 4.0.0' dropdown, and a 'Photon acceleration' checkbox (checked). Below this, the 'Preferred node type' is set to 'Standard\_E4ads\_v6' (32 GB Memory, 4 Cores) and the 'Single node' checkbox is checked (circled in red). The 'Tags' section has a 'Key' and 'Value' field with a trash icon and a link to 'Automatically added tags'. The 'Advanced' section is expanded, showing 'Access mode' with 'Auto' and 'Manual' buttons ( 'Manual' is selected) and a '[Legacy] Single user standard' dropdown. Below this, the 'Single user or group' dropdown is also set to '[Legacy] Single user standard'. At the bottom, there are 'Create' and 'Cancel' buttons.

**Databricks runtime:** Make sure you select the Databricks runtime to 17.3 LTS. AutoML is not available for high versions.

**Cluster:** The cluster is selected as “single node”.

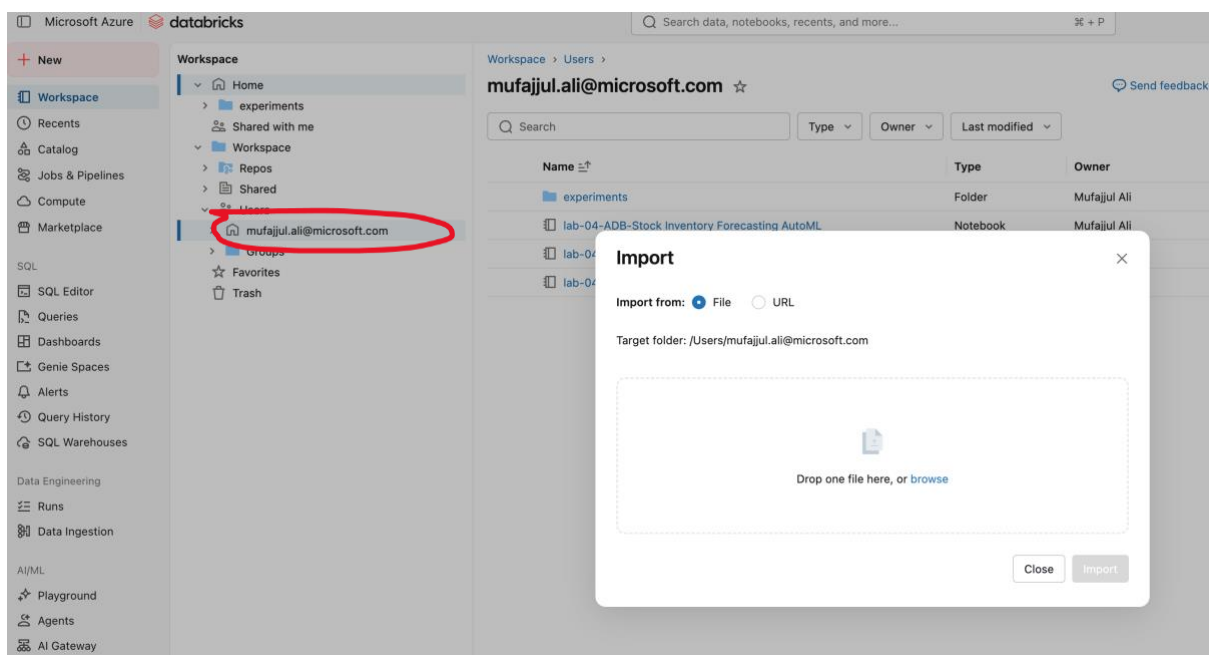
**SKU:** Select Standard\_E4ads\_V6 (recommended), you can also use other types as well, especially if quota is not available.

## Activity 2: Load the notebooks

There are two notebooks used for this lab:

- [data-science/session3-lab4-adb/lab-04-ADB-Stock Inventory Forecasting AutoML \(No UC\).ipynb](#)
- [data-science/session3-lab4-adb/lab-04a-Inventory Forecast Agent with MCP.ipynb](#)

Both labs, need to be imported into the workspace. Please right click on your workspace and select import.



## Activity 3: Run notebook 1

Pease open the notebook, lab-04-ADB-Stock Inventory Forecasting AutoML (No UC).ipynb, and run individual cells.

## Stock Inventory Level Forecasting (No Unity Catalog)

End-to-end ML pipeline for forecasting daily stock inventory levels across multiple SKUs.

| Step | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 1    | <b>Configuration</b> — Set experiment directory and model names                                     |
| 2    | <b>Synthetic Data</b> — Generate daily inventory for 5 SKUs with trend, seasonality & replenishment |
| 3    | <b>EDA</b> — Visualise time series patterns, distribution and seasonal effects                      |
| 4    | <b>Databricks AutoML</b> — Auto-train & tune forecasting models (10-minute budget)                  |
| 5    | <b>Prediction</b> — Load best model, generate 30-day forward forecasts                              |

✓ Yesterday (<1s)

2: Configuration

```
# — Configuration
# Set your own path
EXPERIMENT_DIR = "/Users/mufajjul.ali@microsoft.com/experiments/inventory_forecast"

print(f"MLflow experiment dir : {EXPERIMENT_DIR}")
```

MLflow experiment dir : /Users/mufajjul.ali@microsoft.com/experiments/inventory\_forecast

There is a total of six tasks to be completed here. Please go through each one of them, once you have completed, there should be an endpoint deployed with the best model.

## Activity 4: Run notebook 2

Pease open the notebook, lab-04-Inventory Forecast Agent with MCP.ipynb, and run individual cells.

lab-04a-Inventory Forecast Agent with MCP

lab-04-ADB-Stock Inventory Forecasting A...

File Edit View Run Help Python ☆ Last edit was 2 days ago

Run all

autoML cluster

Schedule

## Inventory Forecast Agent with MCP Server

This notebook demonstrates how to invoke the **inventory-forecast-endpoint** using an AI agent connected via the **Model Context Protocol (MCP)**.

| Step | Description                                                                        |
|------|------------------------------------------------------------------------------------|
| 1    | <b>Install packages</b> — databricks-langchain, mcp, fastmcp                       |
| 2    | <b>Configuration</b> — Endpoint name, workspace URL, LLM endpoint                  |
| 3    | <b>Define MCP Server</b> — FastMCP server exposing the forecast endpoint as a tool |
| 4    | <b>Build Agent</b> — LangGraph ReAct agent connected to the MCP server             |
| 5    | <b>Test Agent</b> — Invoke forecasts via natural language queries                  |

**Architecture:**  
User (natural language) → LangGraph Agent → MCP Server → Model Serving Endpoint → Predictions

✓ Yesterday (30s)

2: Install Required Packages

```
%pip install databricks-langchain langgraph fastmcp mcp -q
dbutils.library.restartPython()
```

There are five tasks to be completed here, please go through each one of them, you should be able to test an agent to use the custom model to perform various asks.

**Note: The deployed model can also be invoked/used by the agents created in Azure Foundry.**

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